Learning the Tie-Line Interface: GNN Fast Screening and Learned Local P/Q Control

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Abstract

Generation interconnection is an information problem. A planner must decide how much power a new plant can inject through a single tie line without breaking voltage stability or thermal limits, and once the plant is connected its smart inverter must keep the grid inside limits using only local measurements and one broadcast signal. We build both ends of that loop on the IEEE 39-bus system with a dedicated gen-tie line under the full AC power-flow equations. A graph neural network screens injection scenarios for congestion and stability headroom, reaching 6.2% RMSE on maximum line loading in distribution, 91% congestion-decision accuracy against 83% for a feature-based MLP, a headroom rank correlation of 0.75, and 0.55 ms per scenario against 256 ms for one AC solve. Under an N-1 topology shift all three surrogates we test fail, at 28 to 32% loading RMSE, which we report as a measured limit rather than a solved problem. A learned local controller maps PCC voltage, real output, and the screening signal to a joint active and reactive dispatch inside the inverter capability curve, and in the closed loop it reduces a congestion-plus-voltage-plus-loss objective by 75% over the IEEE 1547 volt-var baseline. Every number regenerates from saved artifacts. The transferable finding is that a purely local controller cannot see congestion, so the cheapest useful interface for a distributed energy resource is one well-validated scalar broadcast by a network-level model.

1 Introduction

The interconnection queue is the front door of the energy transition, since in the United States more than a terawatt of generating capacity is waiting to connect and the studies that decide whether a plant may connect check thousands of injections against the voltage and thermal limits of the bulk network [Gorman et al., 2025]. FERC Order 2023 exists to make those studies fast enough to re-run as the queue shifts [FERC, 2023], and plants behind weak tie lines are curtailed on the operating side because their real-time output congests the network [Gorman et al., 2025, ERCOT, 2024]. Both situations are one information problem, because the planner at the boundary and the plant at the point of common coupling both need to know what the plant's power is doing to the whole grid.

Two gaps sit between the queue and the control room. The first is screening. Full AC power flow is the gold standard, but one solve takes seconds on a real system and a queue study runs thousands of them, while linearized DC screening is fast and blind to voltage stability, the limit that binds on weak tie lines. The second is control, where the standard today is a static volt-var curve from IEEE 1547-2018 [IEEE, 2018], a fixed function of local voltage that neither sees congestion nor adapts to the plant's headroom. Learned local control exists on distribution feeders [Guo and Lei, 2023, Liu et al., 2025], but a controller is only as good as the signal it conditions on, and a purely local controller cannot know that a distant line is congested.This paper builds the two ends of the loop and measures them together. The first is a graph neural network that fast-screens injection scenarios for a gen-tie, predicting maximum line loading, minimum voltage, the loadability margin λ that

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bounds voltage stability, and the congestion headroom κ that bounds how far plant output can rise before a line reaches its rating, since graph architectures are the established choice for injections that propagate along network edges [Donon et al., 2020, Falk et al., 2019, Zhang et al., 2025]. The second is a small learned policy that maps what the plant can measure, which is PCC voltage and its own output plus one broadcast scalar from the screening module, to a joint active and reactive dispatch, trained to imitate the grid-optimal dispatch of a dense AC scan. Every policy is evaluated with the full AC power flow in the loop, because matching optimal setpoints in isolation does not guarantee that the loop works.

Two measured results carry the paper, and they pull in opposite directions.The screening result comes with an honest negative, since the graph network screens congestion decisions accurately in distribution at 91% against 83% for an MLP on the same features and runs roughly 470 times faster than one AC solve, while under an N-1 topology shift every surrogate we test fails at 28 to 32% loading RMSE. The control result is positive, since the learned policy reduces the congestion-plus-voltage-plus-loss objective by 75% over the IEEE 1547 baseline inside the loop and keeps the improvement under the out-of-distribution load regimes. Failure modes are reported on both ends.

2 Problem formulation

AC power flow and the gen-tie model. A transmission network is a set of buses $i \in \mathcal{B}$ joined by lines and transformers, and the AC power-flow equations balance net active and reactive injection at every bus,

$$P_i = \sum_j V_i V_j (G_{ij} \cos\theta_{ij} + B_{ij} \sin\theta_{ij}), \qquad Q_i = \sum_j V_i V_j (G_{ij} \sin\theta_{ij} - B_{ij} \cos\theta_{ij}), \tag{1}$$

where V_i is the voltage magnitude at bus i, $\theta_{ij} = \theta_i - \theta_j$ is the angle difference, and G_{ij} and B_{ij} are the conductance and susceptance of the connecting line. We solve (1) with the full Newton-Raphson solver of pandapower [Thurner et al., 2018], and a non-converging solve is the practical signature of voltage collapse and is treated as infeasible.

The generation tie is a dedicated line from a plant bus p to a point of common coupling c on the bulk network, and the plant is an inverter-based resource with active output $P_g \in [0, P_{\max}]$ and reactive output constrained by its apparent-power rating,

$$Q_g^2 \le S_{\max}^2 - P_g^2, \tag{2}$$

so the reactive headroom $Q_{\max}(P_g) = \sqrt{S_{\max}^2 - P_g^2}$ shrinks as the plant produces more. Injecting P_g through the tie changes every line flow and every voltage in the network, and the screening question is by how much.

Impact metrics. For a scenario at plant output P_g we compute four quantities from full AC solves. Maximum line loading $L_{\max}$ is the largest line flow as a percent of rating, and a scenario is congested when $L_{\max} > 100$. Minimum voltage $V_{\min}$ is the lowest bus voltage. The loadability margin λ is the largest factor by which loads and plant output can be scaled together before (1) loses a solution, minus one, and congestion headroom κ is the largest factor by which plant output alone can be raised before any line reaches its rating. Both margins are computed by bisection with a full AC solve at every step, so both are expensive and both are what the surrogate has to reproduce.

Information structure for local control. The plant can measure its PCC voltage V_{pcc} and knows its own real-time output P_g, but it cannot observe the bulk grid, so we give it one additional scalar ζ, the predicted maximum line loading from the screening module, which models the way ERCOT-style curtailment signals already broadcast a single congestion indicator [Gorman et al., 2025]. The control problem is to choose the dispatch (P_g, Q_g) inside (2) as a function of $(V_{\text{pcc}}, P_g, \zeta)$ that keeps the grid inside its limits. The static baseline is the IEEE 1547 volt-var curve, with reactive output a fixed function of voltage at a ± 0.012 pu deadband and full capability at $|V_{\text{pcc}} - 1| = 0.06$ [IEEE, 2018].

3 Method

End 1, the graph screener. The network is a graph whose nodes are buses and whose edges are lines and transformers. Node features are per-bus active and reactive load and generation plus flags for the slack, plant, and generator buses, edge features are resistance, reactance, susceptance, rating, tap ratio, and a line-versus-transformer indicator, and the scenario-level plant output and load regime

are appended as two extra node features. A three-layer graph isomorphism network with edge-feature message passing and sum-pooling readout [Xu et al., 2019] predicts $L_{\max}$, $V_{\min}$, λ, and feasibility, trained with mean squared error plus binary cross-entropy, and a second instance on the margin subset predicts κ. Training targets come exclusively from AC power-flow solves, so the model inherits the physics of (1) rather than a linearized approximation, and evaluation is constraint-aware, reporting the accuracy of the induced congestion decision, the rank correlation of the margins, and inference time against one AC solve.

End 2, the learned local controller. The reference dispatch comes from a dense scan that solves the full AC power flow over a grid of Q_g values inside (2) and picks the reactive output minimizing

$$J(P_g, Q_g) = \frac{\max(0, L_{\max} - 100)}{100} + \frac{\max(0, |V_{\mathrm{pcc}} - 1| - 0.012)}{0.1} + 0.5\,\frac{L_{\mathrm{loss}}}{P_{\mathrm{load}}}, \tag{3}$$

which penalizes congestion, voltage deviation, and network losses relative to total load. Each term is dimensionless before it is summed: loading as a fraction of the line rating with 100 as the congestion threshold, voltage deviation in per-unit against a 0.012 deadband scaled by 0.1, and losses as a fraction of total load, so the three contribute comparably. When a scenario is congested at the best reactive-only dispatch we repeat the scan over curtailment levels and keep the jointly optimal pair. A small multilayer perceptron policy π_ϕ maps $(V_{\mathrm{pcc}}, P_g, \zeta)$ to a reactive setpoint Q and a curtailment ratio $r \in [0.2, 1]$, with dispatch $P_g^{\mathrm{new}} = r\,P_g$, trained by regression against the oracle targets. It is then evaluated the way it would deploy, since the plant measures V_{pcc}, the policy returns (Q, r), the power flow is re-solved, and the loop repeats to a fixed point in three iterations, which is the system-level test because a controller that matched setpoints in isolation could still destabilize the loop. Its baselines are the IEEE 1547 volt-var curve and a fixed 0.95 power factor in that same loop, plus the dense-scan oracle.

4 Experiments

Setup. All experiments use the IEEE 39-bus New England system [Athay et al., 1979] from pandapower [Thurner et al., 2018], augmented with a plant bus connected to bus 4 by a 15 km, 230 kV gen-tie line rated at $S_{\max} = 1000$ MVA with $P_{\max} = 900$ MW. Scenarios randomize per-bus loads by $\pm 15\%$ around a regime multiplier, redispatch the nine conventional generators by $\pm 10\%$, and draw plant output uniformly in $[0, 900]$ MW at training regimes $\{0.85, 1.0, 1.15, 1.30\}$ and out-of-distribution regimes $\{1.45, 1.60\}$, with 3,000 screening scenarios (margins by bisection on 250) and 750 control scenarios of which 150 are out of distribution.

Result 1: the graph network screens accurately and cheaply, and fails on a topology shift. We train on 2,400 of the 3,000 scenarios and evaluate on the held-out 600, the out-of-distribution load regimes, and the N-1 topology, over four seeds. The headline number is the accuracy of the induced congestion decision $L_{\max} > 100\%$, which is 91% in distribution against 83% for an MLP on the same node features. Loading itself is predicted at $6.2 \pm 0.3\%$ RMSE in distribution and $10.6 \pm 0.3\%$ under the load shift, and a full scenario screens in 0.55 ms against 256 ms for one AC solve, a factor of roughly 470, which is what makes re-screening a queue feasible. A linear baseline reaches 4.9% RMSE, better than the graph network, because inside the training envelope the loading response is close to linear and the graph structure pays off mainly in the decision accuracy and the margins.
The shift that matters for interconnection is structural rather than distributional, because a line trips and the graph changes while the injections barely move. The N-1 split removes the most heavily loaded line, buses 15 to 18 at 77.9% in the base case, and re-solves every scenario on the modified network, and this is the paper's honest negative. No surrogate generalizes, since the graph network fed the modified graph reaches 31.3% RMSE and the MLP and linear baselines, fed node features that are statistically identical between the two shifts, reach 28.0% and 31.8%. The topology change moves the loading distribution outside the training envelope and all three models underpredict it, and because the failure holds for the graph-structured model too it bounds what surrogate screening can promise without retraining. Voltage-stability headroom is the harder target, and Figure 1b separates the two margins, since the congestion-headroom prediction correlates with the true injection multiplier at Spearman $\rho = 0.75$ pooled over seeds, which ranks which injections are closest to congesting the network, while the loadability-margin prediction averages $\rho = 0.29$ and goes as low as -0.35 on individual seeds. That asymmetry is why the scalar broadcast to the controller is the congestion prediction rather than the stability margin.

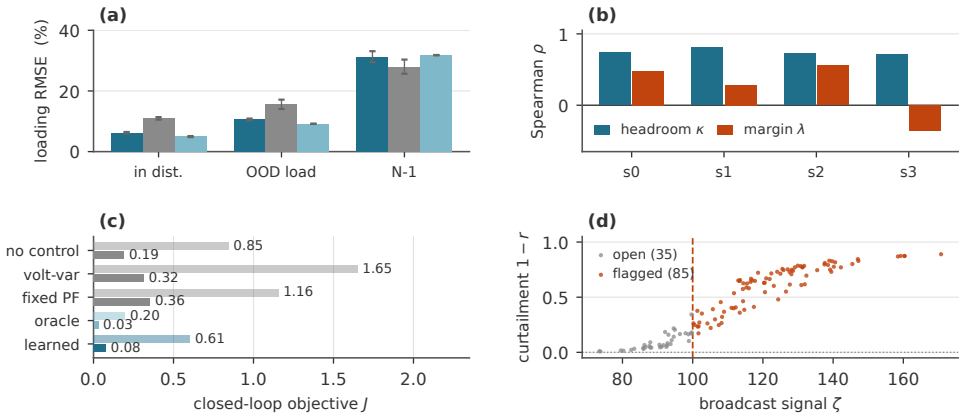


Figure 1: Both ends of the loop under the full AC power flow. Panels (a) and (b) are screening, mean $\pm$ std over four seeds: (a) maximum line loading RMSE on the held-out training regime (in dist.), the out-of-distribution load regimes, and the N-1 topology, where no model survives; (b) rank quality of the two margins per seed, where congestion headroom κ is usable and loadability margin λ is not. Panels (c) and (d) are control, inside the same AC power flow loop iterated to a fixed point: (c) the mean objective (3), and (d) curtailment against the broadcast signal ζ, where the dashed line is the congestion threshold.

Result 2: the learned local controller beats the static curve inside the loop. Trained by regression against the dense-scan oracle on 600 scenarios, the policy is evaluated in the closed loop on 120 held-out plus 150 out-of-distribution scenarios against the IEEE 1547 volt-var curve, the fixed power factor, and the oracle. Figure 1c gives the outcome. On held-out scenarios the learned policy reaches a mean objective $J = 0.079$ against $J = 0.317$ for the volt-var droop, a reduction of 75%, and it keeps the lead out of distribution at $J = 0.605$ against 0.846 for no control. The reduction comes from where the static curve is blind, because the droop reacts only to local voltage while the learned policy conditions on the broadcast signal ζ, so on held-out scenarios it curtails in 85 of 120 cases at a mean of 0.61 when it flags congestion against 0.10 when it does not, concentrating curtailment where the network is stressed. The dense scan achieves $J = 0.031$, so the learned policy closes 83% of the gap between the static baseline and the grid-optimal dispatch.
The congestion rate tracks the same ordering, at 58% of held-out scenarios for both no control and the volt-var droop, 59% for the fixed power factor, 31% for the learned policy, and 22% for the oracle, against 97 to 100% for every policy under the out-of-distribution load regimes.

Failure modes. We report failures alongside means. The headline failure is structural, since all three screening models degrade to 28 to 32% loading RMSE under the N-1 shift, so a surrogate must be re-graphed and re-validated or retrained whenever the topology changes, and flagged scenarios should defer to a full AC solve. The twelve worst held-out errors are high-loading regime-1.30 scenarios the model underpredicts, and the loadability margin is the weak target at $\rho = 0.29$. On the control side the learned policy is worse than the static droop on 15% of held-out scenarios, which are cases where the graph network underpredicts congestion, so a flagged case should get a slower-solver or human review.

5 Discussion and limitations

The two results are two ends of one loop, because the screening module broadcasts a single scalar and the plant's policy turns that scalar plus its local voltage into an action that measurably reduces congestion, voltage deviation, and losses, which is the part that generalizes past this test system. The asymmetry between the two margins is what makes the choice of scalar consequential. The limits are exact. The 39-bus system with one gen-tie is a laboratory grid rather than a deployment, the signal is reliable for congestion and not for stability headroom, surrogate screening does not generalize to the N-1 shift we measure, and the controller is evaluated to a fixed point of the static equations rather than against transient dynamics, so nothing here speaks to angle or frequency stability. The supported pathway is narrow and real, since a screener running in milliseconds lets an operator re-run an interconnection study as the queue changes, with a fallback to the static curve whenever the signal is unreliable.

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Availability of code and data

The scenario generators, trainers, figure script, generated datasets, and per-seed metric records are available from the author on request, together with the exact commands that regenerate every number and every figure in this paper. All power-flow quantities come from pandapower AC solves with no linearized approximation anywhere in the training targets or the evaluation loop, the margins are bisection results over full AC solves, and no proprietary data is used.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to check every quantitative claim against the saved per-seed result files and generated datasets of the accompanying pipeline, to confirm that each figure asserts the headline values it plots, and to verify citations against their sources. The authors directed the work and take responsibility for all content.